Mathematics

An Introduction to Numbers, Algebra, Geometry and Formulas

Mathematics is a branch of science that deals with numbers, quantities, shapes, patterns, and calculations. It helps us solve problems, measure objects, calculate distances, manage money, and understand the world around us.

1. WHAT IS MATHEMATICS

Mathematics is used in almost every part of our daily lives.

Examples include:

- Calculating shopping bills.
- Measuring the length and width of a room.
- Calculating time and distance.
- Managing money and budgets.
- Calculating percentages and discounts.
- Designing buildings and bridges.

2. Basic Mathematical Formulas

Mathematical Concept	Formula
Addition	$a + b = c$
Subtraction	$a - b = c$
Multiplication	$a \times b = c$
Division	$a \div b = c$
Percentage	$\frac{Value}{Total} \times 100$
Average	$\frac{Sum\ of\ Values}{Number\ of\ Values}$
Simple Interest	$SI = \frac{P \times R \times T}{100}$
Speed	$S = \frac{Distance}{Time}$

2. Algebra

Algebra is a branch of mathematics that uses letters and symbols to represent unknown numbers.

Example:

$$x + 5 = 15$$
$$x = 15 - 5$$
$$\boxed{x = 10}$$

Here, x represents an unknown number.

Important Algebraic Formulas:

1. **Square of a sum:**

$$(a+b)^2 = a^2 + 2ab + b^2$$

2. **Square of a difference:**

$$(a-b)^2 = a^2 - 2ab + b^2$$

3. **Difference of two squares:**

$$a^2 - b^2 = (a-b)(a+b)$$

4. **Geometry**

Geometry is the branch of mathematics that studies shapes, sizes, angles, lines, and measurements**.**

Important Geometry Formulas:

Shape	Formula
Square Area	$A = a^2$
Square Perimeter	$P = 4a$
Rectangle Area	$A = l \times w$
Rectangle Perimeter	$P = 2(l + w)$
Triangle Area	$A = \frac{1}{2}bh$
Circle Area	$A = \pi r^2$
Circle Circumference	$C = 2\pi r$

Perimeter and Circumference - A Plus Topper

Area of Different Shapes (with Formula) - Mensuration Class 8 - Worksheets Library

Areas of Composite Shapes

5. Percentage and Profit & Loss

Percentage represents a number as a fraction of 100.

Percentage Formula:

$$Percentage = \frac{Obtained\ Marks}{Total\ Marks} \times 100$$

Example:

A student gets 80 marks out of 100.

$$\frac{80}{100} \times 100 = 80\%$$

Profit For

$$Profit = Selling\ Price - Cost\ Price$$

Loss Formula:

$$Loss = Cost\ Price - Selling\ Price$$

Discount Formula:

$$Discount = \frac{Discount\ Rate \times Original\ Price}{100}$$

6. Statistics

Statistics is the branch of mathematics that deals with collecting, organizing, analyzing, and interpreting data.

Important formulas:

Mean (Average):

$$Mean = \frac{\sum x}{n}$$

Median: The middle value when numbers are arranged in order.

Mode: The number that appears most frequently.

Example:

Numbers: 2, 4, 4, 6, 9

- Mean = 5
- Median = 4
- Mode = 4

7**. Why Is Mathematics Important**?

✓ Mathematics improves logical thinking.

✓ It develops problem-solving skills.

✓ It is important in computer programming.

✓ It helps in business and financial management.

✓ It is used in engineering and construction.

✓ It helps scientists perform experiments.

✓ It is useful in everyday calculations.

CONCLUSION

Mathematics is an essential subject that helps us understand numbers, patterns, shapes, and calculations. Mathematical formulas make it easier to solve complex problems in algebra, geometry, statistics, and everyday life. Learning mathematics develops logical thinking and prepares us for science, technology, engineering, and business.

نوٹ: تمام سوالات کے جوابات دیجیے۔

سوال نمبر 1: مندرجہ ذیل اشعار میں سے کسی دو کی تشریح کیجیے۔ 2x3=6

بار بار اس کے در پہ جاتا ہوں
حالت اب اضطراب کی سی ہے

آدمی آدمی سے ملتا ہے
دل مگر کم کسی سے ملتا ہے۔

وہ جانِ جاں نہیں آتا تو موت ہی آتی
دل و جگر کو کہاں تک بھلا سنبھالا جائے۔

۔ رنج سے خوگر ہوا انسان تو مٹ جاتا ہے رنج
مشکلیں مجھ پر پڑیں اتنی کہ آساں ہ

سوال نمبر 2: متن کے حوالے سے کسی دو کی تشریح کیجیے۔

بھائی! اب میں نے فیصلہ کیا ہے کہ وہ میرے ساتھ اس گھر میں نہ رہیں گے۔

غالب کے کلام سے ... ہو جائیں تو پکنک دینے کے قابل، گھر کے ... اور کمینوں کی حفاظت کریں۔

اقبال! تم ہی بتاؤ مرزا صاحب کہ اس دل کے سفر میں بے نصیب دشمن، جان جو ہم خرچ نہیں ہے

یہ تو واضح ہوتی نہیں سکتا کہ ہم پروفیسر کو سرہائیں۔

سوال نمبر 3: ذیل میں سے کسی ایک نظم کا خلاصہ لکھیے۔